Massey University
College of Sciences

161.304: Computer Laboratory 2

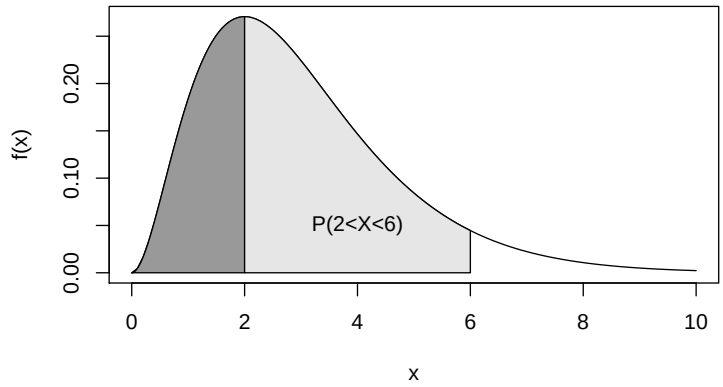
Exercise 1: Some probability questions

Do Practice activities 2E1-2E4 of the *Statistical Rethinking* book.

Exercise 2: Probability Calculations
Probabilities for interval ranges can be computed conveniently using the *cumulative distribution function*, $F(x) = \mathsf{P}(X \leq x)$ for a random variable X. In particular, to compute $\mathsf{P}(a < X \leq b)$, note that

$$\begin{aligned} \mathsf{P}(a < X \leq b) &= \mathsf{P}(X \leq b) - \mathsf{P}(X \leq a) \\ &= F(b) - F(a) \end{aligned}$$

The figure below illustrates why this is he case for a continuous random variable. The two grey areas together are $\mathsf{P}(X \leq 6)$, while the dark grey area corresponds to $\mathsf{P}(X \leq 2)$. It follows that the light grey area provides $\mathsf{P}(2 < X \leq 6)$.



Recall that the R functions for computing cumulative probabilities have names of the form `pdist` where `dist` is replaced by the requisite R distribution name.

Calculate the probabilities below using R.

1. $\mathsf{P}(X = 0)$ where $X \sim \mathsf{Pois}(3)$.
2. $\mathsf{P}(X = 2)$ where $X \sim \mathsf{Pois}(4.321)$.
3. $\mathsf{P}(X \leq 4)$ where $X \sim \mathsf{Pois}(3)$.
4. $\mathsf{P}(X < 4)$ where $X \sim \mathsf{Pois}(3)$.
5. $\mathsf{P}(2 < X \leq 4)$ where $X \sim \mathsf{Pois}(3)$.
6. $\mathsf{P}(X \leq 5)$ where $X \sim \mathsf{N}(4, 1.5^2)$.
7. $\mathsf{P}(X > 5)$ where $X \sim \mathsf{N}(4, 1.5^2)$.
8. $\mathsf{P}(2.5 < X \leq 6.5)$ where $X \sim \mathsf{N}(4, 1.5^2)$.

9. $\mathsf{P}(X < 1.4)$ where $X \sim \mathsf{Exp}(1.2)$.
10. $\mathsf{P}(3.21 < X < 6.54)$ where $X \sim \mathsf{Gam}(3, 1.1)$.

Exercise 3: Plotting Binomial Probability Mass Functions

Produce a plot of the mass function for the $\mathsf{Bin}(n, p)$ distribution with $n = 10, p = 0.6$, using dots to mark the probabilities. You will need to use the R command `dbinom` (for which you can use the help function to learn the syntax)

Exercise 4: Plotting Density Functions
Using Example 3.28 from notes and/or Example 26 from lectures as a template, produce plots of the following probability density functions. In each case, experiment with the range of x-values so that you get a plot that covers a suitable range for the density in question.

1. The density function for the $\mathsf{N}(3, 0.5^2)$ distribution.
2. The density function for the $\mathsf{Exp}(0.2)$ distribution.
3. The density functions for $\mathsf{Gam}(1.5, 2)$ and $\mathsf{Gam}(3, 2)$ distributions. Plot these in the same figure, using a black line for the first and a red line for the second. Note that the argument `col=2` (or `col="red"`) to the `plot` or `lines` function will produce a line in red. Add an appropriate legend to the plot.